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I am a researcher in the field of Computer Vision, AI, Optical Fiber Sensor, and Instrumentation Electronics Development. I am known as a person who is committed to working as a collaborative and positive team member. I am used to working in large teams with interdisciplinary backgrounds. I am curious and enjoy learning something new. I enjoy looking at other possibilities for learning outside my area of expertise.

EDUCATION

UNIVERSITY OF TECHNOLOGY MALAYSIA

DOCTOR OF PHILOSOPHY

Malaysia

- PhD in Electrical Engineering Majoring in Computer Vision and Image Processing **September 2026 – Now**
- Awardee UTM ASCEND Bright Talent International Scholarship

UNIVERSITY OF TECHNOLOGY MALAYSIA

MASTER OF PHILOSOPHY

Johor Bahru, Malaysia

October 2024 – June 2026

- Master of Philosophy in Department of Mechatronics Engineering Majoring in Control System and Optoelectronics
- Graduate Lecture Assistant at Photonics Research Center
- Silver Medal in New Academia Learning Innovation (NALI) 2025 – University of Technology Malaysia
- Team Members in UTM Rocketry (Avionics Department)

UNIVERSITAS NEGERI SEMARANG

BACHELOR'S DEGREE

Semarang, Central Java

August 2020 – December 2024

- CGPA 3.55 / 4.00
- Bachelor of Science in Instrumentation Electronics Physics (ELINS)
- 1st Place in LKTIN Competition at Bangka Belitung University (September 2023)
- 1st Place in ESSAY Writing National Competition at Universitas Negeri Semarang (October 2023)
- 3rd Place in ESSAY Writing National Competition at Universitas Negeri Semarang (October 2022)
- Grand Finalist in Scientific Writing National Competition at Lampung State University (August 2022)
- Team Member in Indonesian Flying Robot Contest 2021 (KRTI 2021) – Electronics and Design Department
- Team Member in Indonesian Flying Robot Contest 2023 (KRTI 2023) – Electronics and Design Department

RESEARCH, INTERN, AND WORK EXPERIENCE

PARTICIPANT AI IN MEDICINE SUMMER SCHOOL AND BOOTCAMP 2026

Kuala Lumpur, Malaysia

- Participant In AI in Medicine Summer School and Bootcamp in University Malaya **27 July – 5 August 2026**

ROCKETRY TEAM UNIVERSITY OF TECHNOLOGY MALAYSIA

Johor Bahru, Malaysia

- PCB Designer and CAD in Avionics Rocket Part **August 2025– Now**

GUEST LECTURE IN UNIVERSITY OF BANGKA BELITUNG

Bangka Belitung, Indonesia

February 2025 and November 2025

- Guest Lecture in the Department of Physics with the topic “Deep learning applications for MMI-Based optical sensors”
- Guest Lecture in the Faculty of Engineering and Science with the topic “Implementation of Machine Learning Methods for Heavy Metal Detection in Former Tin Mining Waters using Smart Electronic Sensors”

NATIONAL RESEARCH AND INNOVATION AGENCY REPUBLIC INDONESIA (BRIN)**Serpong, Indonesia**

- Research Assistant at Research Center for Photonics
- Development of Smart Weapon by Using IR Lens for Scope and Object Detection Algorithm based-on YOLOv11 - Current Research

September 2024 – August 2026**PT. BORINE TECHNOLOGY INDONESIA****Semarang, Indonesia****RnD Engineer in PCB Department****April 2024 – August 2024****DEPARTMENT OF PHYSICS, UNIVERSITAS NEGERI SEMARANG****Semarang, Indonesia**

- Research Assistant Lecture "Photonic Research Group", January 2022 – March 2024
- Assistant Lecture in Course Microcontroller, August 2022 – September 2023
- Assistant Lecture in Course Instrumentation System Programming, August 2022 – September 2023
- Assistant Lecture in Course Computer Programming, January 2024 – April 2024

2022 -2024**UNIVERSITI TEKNOLOGI MALAYSIA – FACULTY OF ELECTRICAL ENGINEERING****Johor Bahru, Malaysia****Intern (September 2023)****2023**

- Simulating A Waveguide-Based Glucose Sensor with Surface Plasmon Resonance (SPR) Method Using Beamprop

NATIONAL RESEARCH AND INNOVATION AGENCY REPUBLIC INDONESIA**Serpong, Banten****Student Intern (28 June 2023 – 04 September 2023)****2023**

- Arduino Nano Microcontroller-Based Optical Power Meter Development

PT. ALITA & INSTITUT TEKNOLOGI SEPULUH NOVENBER (ITS)**West Jakarta, DKI Jakarta****Student Intern (01 August 2021 – 08 December 2021)****2021**

- GPS Module-Based Delivery Drone Using ESP8266

SKILLS**BASICS SKILLS:**

- Python
- C++
- MATLAB
- Assembly Language
- Pascal

HARDWARE SKILLS:

- Assembly Electronics Circuit, Design, and Wiring
- Optical Fiber Sensor Fabrication
- CNC Machine

SOFTWARE SKILLS:

- Microsoft Office (Word, Excel, Power Point)
- Corel Draw
- KiCAD
- ZW3DCAD
- Autodesk Inventor
- ANSYS Lumerical FDTD
- COMSOL Multiphysics
- Beamprop

DESIGN AND ANALYSIS SKILLS:

- 3D Design, Simulation, and Analysis
- PCB Design
- Photonics Analysis & Simulation

LANGUAGES:

- Indonesia (Native Language)
- Hakka (Native Language)
- English (Professional)
- Mandarin (Fluent)

PUBLICATIONS & PATENT

- Development of Color Detection Based on Camera Infrared Filter with Python Image Processing (**IEEE Scopus, 2024 – Best Paper Award**)
- Optimized selectivity in no-core fiber sensors for high refractive index detection (**WoS Q2, 2025**)
- Optimizing Convolutional Layer in YOLOv8: Enhancing Accuracy and Efficiency for Camouflaged Object Detection in Complex Environments (**IEEE Scopus, 2025**)
- Effect of Surrounding Refractive Index Changes on Amplitude Response in MMI-Based Fiber Optic Sensors (**IEEE Scopus, 2025**)
- Design and Characterization of a Simple Temperature Sensor based-on a Polymer Sine S-Bend Optical Waveguide Structure (**WoS Q2, 2025**)
- Fabrication and Characterization of Low-Cost Optical Waveguides Using Recycled Expanded Polystyrene (EPS) as Core Material (**WoS Q2, 2025**)
- Comparative Analysis of Machine Learning Algorithms for Wide-range Refractive Index Classification in Multimode Interference No-Core Fiber Sensors (**Conference Paper, 2026**)
- Quantized LSTM For Predictive Thermal Monitoring Temperature of Lithium-Ion Battery (**WoS Q3, 2026**)
- PySpecTrace: A Python-based Graphical User Interface for Real-Time Spectroscopy Analysis and Data Analysis (**WoS Q1, 2026**)
- Machine Learning-Based Wide-Range Refractive Index Sensing with Synthesis-Tuned Nanoparticles Coating Layer on MMI Fiber Sensors (**WoS Q2, 2026**)
- Design and Characterization of Low Cost-Visible Light Optical Power Meter (LC-VLOPM) (**Patent, 2023**)
- Code for Optical Power Meter Measurement (**Patent, 2023**)
- Augmented Segmentation Utility (ASU) for Object Segmentation Software (**Copyright, 2026**)
- SegLab v1.0 (Segmentation Lab for Medical Image) Software (**Copyright, 2026**)
- Optical Spectrum Analyzer Utility Software (**Copyright, 2026**)